Physical activity (PA) has pervasive health benefits and is genetically influenced. There is also a wide range of interindividual differences in requirements for PA in daily life. Some jobs require much more physical exertion than others; some individuals must exert more effort in their households than others; some individuals experience most of their PA while at leisure. PA related to work and home-time tend to differ phenomenologically from exercise taken at leisure, occurring more in concentrated bursts in the first two cases, and with the potential to be more sustained in the latter case. Using data from the Million Veteran Program (MVP), we conducted genetic analyses of leisure, work, and home-time PA. For leisure, for individuals of European (EUR) ancestry, n = 189,812 and SNP-based heritability ($h^2$) = 0.083 ± 0.005; for African ancestry, n = 27,044; $h^2$ = 0.034 ± 0.017; and for Latin American ancestry, n = 10,263; $h^2$ = 0.083 ± 0.036. EUR and cross-ancestry meta-analyses with UK Biobank identified 67 and 70 lead variants, respectively. Leisure-time PA was genetically distinct from PA at home or work, with the latter two showing weaker associations with health benefit on pathologies and lifespan. Mendelian randomization analyses showed protective effects of leisure-time PA on cardiovascular and respiratory system diseases, metabolic traits, aging, and other traits. There was also a protective role of leisure-time PA against COVID-19 hospitalization. Further analyses helped to delineate the relationships between PA and a range of psychiatric traits. These findings provide new insights into the biology of PA, showing the causal health benefits of leisure-time PA.

**Keywords** Genomics, GWAS, Physical activity

**Consortia**
VA Million Veteran Program.

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## Epigenome-Wide Analyses of Pubertal Development in Adolescence: Connections with Biological Aging

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**Abstract**
Puberty is highly heritable and responsive to environmental exposures and signals. Earlier and faster puberty is related to early life adversity and worse health across the lifecourse. Novel DNA-methylation methods provide the opportunity to increase biological insights about the etiology of differences in puberty. We leverage salivary DNA-methylation samples from two diverse population cohorts of adolescents to (1) identify genomic regions that are associated with pubertal development and (2) generate a novel DNA-methylation biomarker of pubertal development. We conduct epigenome-wide association studies (EWAS) of precocious puberty, puberty, and pace of puberty in the Texas Twin Project (TTP; n ∼ 1200) and the Future of Families Child Wellbeing Study (FFCWS; n ∼ 2000). We employ these results to create a novel elastic-net algorithm of puberty in TTP and test its performance in other developmental cohorts. We explore the relationship between puberty-related differences in DNA-methylation and insights from the study of biological aging in midlife and older adulthood using targeted enrichment analysis and by relating our puberty methylation profile score (MPS) to accelerated biological aging and pace of biological aging MPSs. DNA-methylation measurements of puberty have the potential to improve health forecasting and accelerate research on development and aging.

**Keywords** DNA-methylation, Epigenetics, Puberty, Development, Aging

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## Parent–Offspring Stress to Externalizing Pathways: A Test of Genetic Mediation Through Negative Emotionality

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**Abstract**
Negative emotionality has been found to underlie continuity in stress and psychopathology, but less is known about whether it drives intergenerational transmission. Using data from a parent–offspring adoption study, we sought to elucidate genetic mediation of parent–offspring stress to externalizing pathways through negative emotionality and whether parent emotional and verbal responsivity mitigated genetic risk. Specifically, this study examined whether the association between birth mother life stress and adoptee externalizing at age 11 was mediated by birth mother negative emotionality ($n$ = 561 children and their birth and adoptive parents) and whether adoptive parent responsivity moderated paths from birth mother life stress and negative emotionality to adoptee externalizing. Birth mother life stress was positively associated with her negative emotionality (β = 0.15, $p$ = 0.010), and birth mother negative emotionality was positively associated with adoptee externalizing (β = 2.38, $p$ = 0.001). The total effect of birth mother life stress was